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4. Increase in temperature, favours the endothermic reaction,

Given that formation of NH_3 is exothermic ie., the reverse reaction is endothermic.

$\therefore$ increase in temperature, shift the equilibrium to left option (c)

5. $H_2O(l) + CO_2(g) \rightleftharpoons H_2CO_3(l)$

(2 mole) (1 mole)

increase in pressure, favours the forward reaction.

option (a)

6. option (a) : wrong statement

Correct statement is, for a system at equilibrium, $Q = K_{eq}$

7. $K_1 = \frac{[NO]^2}{[N_2][O_2]}$ $K_2 = \frac{[NO]^2}{[NO]^2[O_2]}$ $K = \frac{[N_2]^{1/2}[O_2]}{[NO_2]}$

$$\sqrt{K_1} = \frac{[NO]}{[N_2]^{1/2}[O_2]^{1/2}} \quad \sqrt{K_2} = \frac{[NO_2]}{[NO][O_2]^{1/2}}$$

$$\sqrt{K_1.K_2} = \frac{[NO_2]}{[N_2]^{1/2}[O_2]}$$

$\therefore$ $K = \frac{1}{\sqrt{K_1K_2}}$ (option (a)

8. $[A] = 1 \times 10^{-4}$ M ; $[B] = 2 \times 10^{-3}$ M
$[C] = 1.5 \times 10^{-4}$ M

$2A(g) \rightleftharpoons 2B(g) + C_2(g)$

$$K = \frac{[B]^2[C_2]}{[A]^2} = \frac{(2\times10^{-3})^2(1.5\times10^{-4})}{(1\times10^{-4})^2}$$

$$= 6.0\times10^{-2} = 0.06 \text{ (option (a)}$$

$$K_C = \frac{[Products]}{[Reactants]}$$

$$3.2\times10^{-6} = \frac{[Products]}{[Reactants]}$$

$K_C < 10^{-3}$; indicates

that [Reactant] $\gg$ [Product]

option (b)is correct,largely towards reverse direction.

10. for the reaction,

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$;

$\Delta ng = 2 - 4 = -2$

$\therefore$ $K_P = K_C(RT)^{-2}$

$\Rightarrow$ $\frac{K_C}{K_P} = (RT)^2$ option (d)

11.

	AB	A	B
Initial no. of moles	100	-	-
No. of moles dissociated and formed	20	20	20
No. of moles at equilibrium	80	20	20

Total no. of moles at equilibrium = 80 + 20 + 20 = 120

$$K_P = \frac{P_A \cdot P_B}{P_{AB}} = \frac{\left(\frac{20}{120}\times P\right)\left(\frac{20}{120}\times P\right)}{\left(\frac{80}{120}\times P\right)} = \frac{P}{24}$$

$24K_P = P$ (option (a))

12. For reaction given in option (a), (b) & (c) $\Delta ng = 0$

For option (d) $\Delta ng = 2 - 1 = 1$

$\therefore$ $K_P = K_C(RT)$

13.

	PCl_5	PCl_3	Cl_2
Initial no. of moles	0.5	–	–
No. of moles dissociated	x	–	–
No. of moles at equilibrium	0.5 – x	x	x

Total no. of moles at equilibrium =0.5–x + x + x

= 0.5 + x

option (b)